

ARB12: Addressing Geographic and Semantic Bias in Urban Street-View Segmentation for Arab Cities

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European Workshop on Visual Information Processing (EUVIP 2026)



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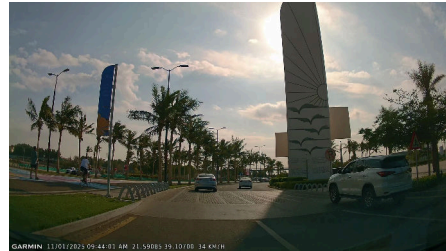
- 1 Introduction
 - Background and Motivation
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 - ARB Saudi Dataset
 - Taxonomy and Labelling
 - Evaluation Protocol
- 4 Results and Discussion

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Street-View Segmentation is not Geographically Neutral

- Street-view segmentation supports walkability, comfort and street-vitality analysis.
- Major benchmarks such as Cityscapes, CamVid and Mapillary Vistas largely reflect European urban scenes.
- Arab streets often combine active ground-floor uses with residential or inactive upper floors.
- Human-scale elements such as shading, benches and kiosks are weakly represented or absent from standard label spaces.



Jeddah, eye-level dashcam frame

Consequence

Planning-relevant features may be misclassified or omitted, biasing downstream urban indicators.

Research Gap and Contributions

Two entangled failure modes

(i) Geographic appearance shift

Lighting, materials, vegetation, road furniture

(ii) Semantic mismatch

Missing planning-relevant concepts

Research gap

No widely adopted benchmark captures human-scale Arab-world street semantics or clearly separates geographic from semantic failure.

1. ARB12 taxonomy

12 region-aware classes, including *Services*, *Streetscape Asset* and *Water*.

2. Eye-level labelling

Ground-floor function separates *Services* from *Structure*.

3. Evaluation protocol

Shared vs full evaluation with boundary-sensitive metrics.

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Related Work



Urban design perspective

AI-generated illustration; visual context only.



Semantic segmentation perspective

Source: Cityscapes Dataset; Cordts et al. (2016).

Segmentation architectures

CNN: FCN, DeepLabV3+, HRNet, UPerNet.
Transformers: SegFormer, Mask2Former, EoMT.

Cross-region transfer

Performance degrades under geographic domain shift, while fixed label spaces compound the problem.

Benchmarks and regional datasets

Cityscapes, Mapillary Vistas and BDD100K do not explicitly target Arab-world streetscapes.

Urban design perspective

Ground-floor activity and small-scale amenities shape pedestrian comfort and street vitality.

Gap: Geographic adaptation is better studied than planning-relevant semantic mismatch.

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ARB Saudi Street-View Dataset

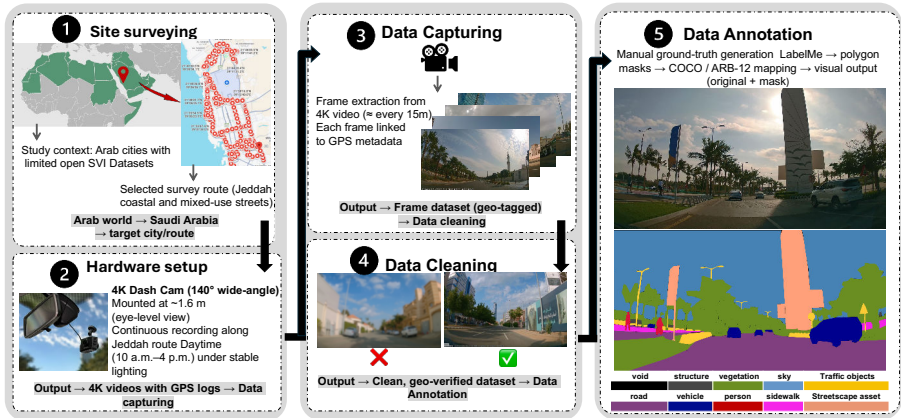
1,238
georeferenced
4K videos

20.6 h
continuous
recording

239 GB
raw
footage

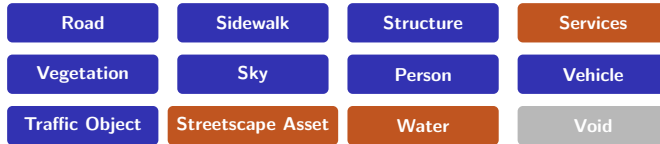
10
Jeddah
neighbourhoods

~1.6 m
camera height
(eye level, 140°)



Ethics: Jeddah Municipality approval; identifiable content anonymised before processing.

The ARB12 Taxonomy



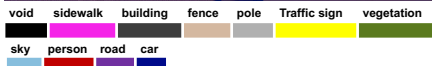
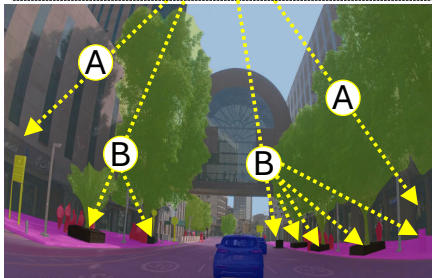
Dark blue = shared with / mapped from Cityscapes. **Orange** = ARB12-specific, absent from Cityscapes. **Grey** = Void, excluded from metric computation.

Services	Streetscape Asset	Water
Ground-floor shops, cafés, pharmacies and civic buildings (mosques, hospitals, hotels).	Benches, litter bins, advertisement boards and street furniture.	Sea, rivers and lakes.

The Eye-Level Labelling Principle

Cityscapes 19 annotation

- A.** merges ground-floor activity into building
- B.** no separate label for non-traffic eye-level elements (merged into background classes)

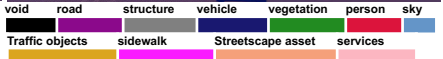
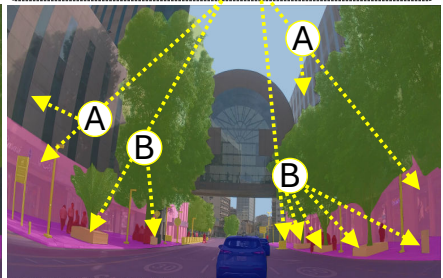


Active ground floor → Services

ARB12 annotation

The Eye-Level Rule:

- A.** ground = services, upper = structure
- B.** non-traffic eye-level elements = streetscape asset



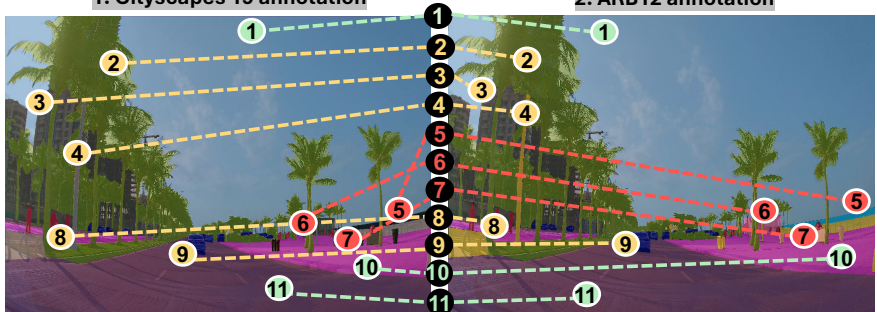
Inactive/residential → Structure

Upper storeys → Structure

Mapping Cityscapes to ARB12 and Metrics

1. Cityscapes 19 annotation

2. ARB12 annotation



1. Sky → Sky	4. Pole/Fence/Sign → Traffic objects	7. Unlabeled → Asset	10. Sidewalk → Sidewalk
2. Vegetation/Terrain → Vegetation	5. Unlabeled → Water	8. Person/Rider → Person	11. Road → Road
3. Building/Wall → Structure	6. Unlabeled → Services	9. Car/Truck/Bus/Train/Motorcycle/Bicycle → Vehicle	

Green: one-to-one

Orange: merged classes

Red: ARB12-specific; absent from Cityscapes

SmIoU / SbloU

Shared classes only.

FmIoU / FbloU

All non-void classes.

Boundary IoU

Boundary accuracy.

Shared-Full → semantic mismatch

Region-Boundary → geometric degradation

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Zero-Shot Transfer to the ARB Dataset

Model	FmIoU	SmIoU	FbIoU	SbIoU	PA	mAcc
FCN (2015)	0.272	0.384	0.061	0.085	0.649	0.402
PSPNet (2017)	0.305	0.433	0.066	0.091	0.790	0.411
DeepLabV3+ (2018)	0.333	0.472	0.083	0.115	0.819	0.437
UPerNet (2018)	0.301	0.429	0.072	0.099	0.815	0.412
HRNet (2019)	0.362	0.516	0.098	0.135	0.856	0.465
SegFormer (2021)	<u>0.413</u>	<u>0.595</u>	<u>0.106</u>	<u>0.145</u>	<u>0.895</u>	<u>0.505</u>
Mask2Former (2022)	0.351	0.500	0.058	0.079	0.857	0.450
EoMT (2025)	0.470	0.676	0.142	0.196	0.909	0.561

SmIoU 0.676 → FmIoU 0.470

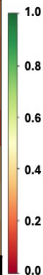
~30% drop with ARB12-specific classes.

PA 0.909 vs SbIoU 0.196

High PA masks poor boundary quality.

Class-Level Behaviour on ARB

FCN	0.62	0.21	0.33	0.70	0.15	0.66	0.15	0.26
PSPNet	0.59	0.27	0.38	0.73	0.15	0.90	0.21	0.23
DeepLabV3+	0.66	0.32	0.51	0.70	0.16	0.93	0.23	0.26
UPerNet	0.61	0.36	0.35	0.71	0.10	0.94	0.21	0.16
HRNet	0.81	0.43	0.50	0.72	0.15	0.95	0.24	0.34
SegFormer	0.89	0.57	0.64	0.79	0.08	0.97	0.41	0.41
Mask2Former	0.75	0.49	0.43	0.75	0.07	0.96	0.20	0.34
EoMT	0.93	0.68	0.75	0.79	0.27	0.98	0.49	0.53
	road	structure	vehicle	vegetation	person	sky	Traffic objects	sidewalk



Shared classes drive SmIoU

EoMT: Road 0.93, Sky 0.98, Vehicle 0.75.
Lower on Person 0.27 and Sidewalk 0.53.

ARB12-specific classes are missing

No valid predictions for Services, Streetscape Asset or Water.

Boundary Quality on ARB

FCN	0.12	0.04	0.09	0.11	0.05	0.13	0.08	0.06
PSPNet	0.13	0.04	0.08	0.10	0.06	0.17	0.08	0.06
DeepLabV3+	0.14	0.06	0.12	0.11	0.05	0.24	0.10	0.08
UPerNet	0.12	0.06	0.09	0.11	0.03	0.24	0.10	0.05
HRNet	0.20	0.08	0.15	0.13	0.05	0.24	0.14	0.09
SegFormer	0.23	0.10	0.15	0.13	0.02	0.28	0.15	0.09
Mask2Former	0.13	0.06	0.06	0.09	0.01	0.20	0.04	0.04
EoMT	0.21	0.15	0.24	0.16	0.11	0.31	0.27	0.11
	road	structure	vehicle	vegetation	person	sky	Traffic objects	sidewalk

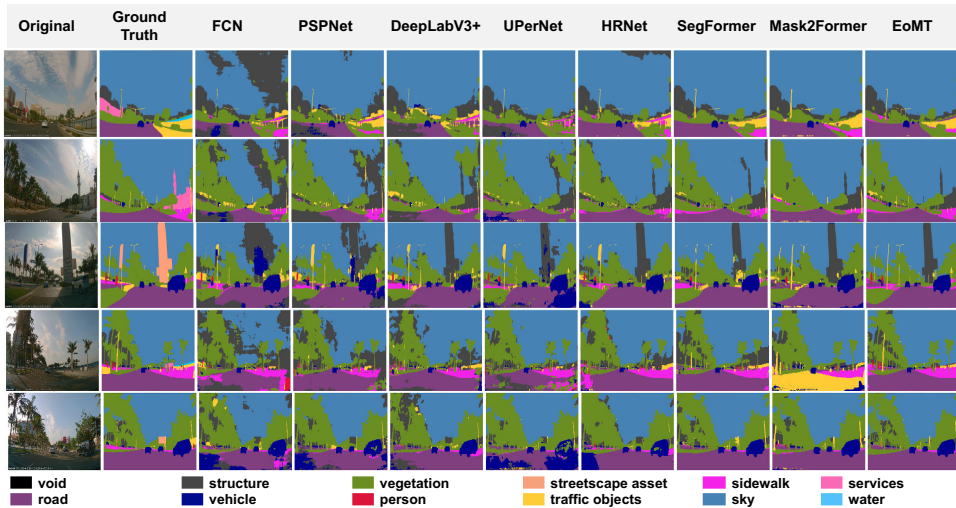
Boundary quality drops sharply

Boundary IoU is consistently lower than region IoU across all models.

Thin objects degrade most

Poles, signs and kerbs show the weakest geometric precision.

Qualitative Evidence on Jeddah Scenes



What this Means for Urban Analytics

1. Indicators inherit the bias

Active frontage, comfort and walkability scores can be underestimated when ARB12-specific classes are missing.

2. Pixel Accuracy can be misleading

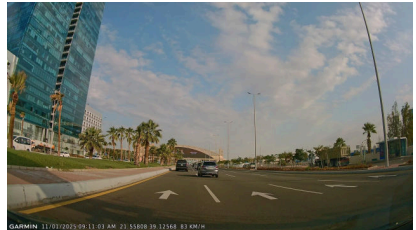
High PA may conceal poor boundary quality and errors on thin, planning-relevant elements.

3. Critical scenes remain challenging

Dense, mixed-use streets with active frontages show lower segmentation reliability.

4. Adaptation must include semantics

Appearance adaptation alone cannot recover concepts absent from the label space.



Representative Jeddah street scenes

Conclusion and Future Work

Summary:

- 1 ARB12: a region-aware 12-class taxonomy with an eye-level labelling principle for Arab-world streetscapes.
- 2 An evaluation protocol separating semantic mismatch from geographic generalisation, with boundary-sensitive metrics.
- 3 Cityscapes-pretrained models hold up on shared classes but degrade sharply on region-specific and human-scale elements - confirming systematic geographic *and* semantic bias.

Future work

Semi-automatic labelling to cut annotation time; extending coverage across Arab cities; domain adaptation and segmentation foundation models for cross-region transfer.



github.com/mawady/euvip26

Taxonomy definition, Cityscapes
mapping, evaluation protocol and
reproduction scripts



Questions?

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Source: Magnific, "Organic Flat Illustration: People Asking

Questions."

References

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- [9] J. Gehl, *Cities for People*. Island Press, 2013.

Appendix — In-Domain Reference on Cityscapes *val*

Model	FmIoU	FbIoU	PA	mAcc
FCN (2015)	0.460	0.304	0.956	0.509
PSPNet (2017)	0.454	0.314	0.959	0.502
DeepLabV3+ (2018)	0.465	<u>0.323</u>	0.963	0.511
UPerNet (2018)	0.456	0.319	0.960	0.500
HRNet (2019)	<u>0.473</u>	0.337	<u>0.965</u>	<u>0.518</u>
SegFormer (2021)	0.430	0.280	0.958	0.479
Mask2Former (2022)	0.374	0.200	0.938	0.426
EoMT (2025)	0.491	0.311	0.969	0.536

Why this matters

Evaluated in ARB12 space, so the values are *not* comparable with official 19-class Cityscapes scores. The three ARB12-specific classes score zero in-domain as well — the mismatch is in the label space, not in the Saudi imagery.

Appendix — Accuracy–Efficiency Trade-off

Model	SmIoU	Time (s/img)
FCN (2015)	0.384	4.2
PSPNet (2017)	0.434	4.5
DeepLabV3+ (2018)	0.472	5.0
UPerNet (2018)	0.429	8.0
HRNet (2019)	0.517	4.2
SegFormer (2021)	0.595	1.3
Mask2Former (2022)	0.500	1.8
EoMT (2025)	0.676	6.5

At city scale

SegFormer is five times faster than EoMT for 88% of its shared accuracy; HRNet sits in between. None of these choices rescues the ARB12-specific classes.

Appendix — Boundary IoU Definition

For class c , with predicted and ground-truth masks P_c, G_c :

$$\text{B-IoU}_c = \frac{|(P_c \cap B_c) \cap (G_c \cap B_c)|}{|(P_c \cap B_c) \cup (G_c \cap B_c)|}, \quad B_c = \text{dilate}(\partial P_c \cup \partial G_c, \tau)$$

$$\tau = \max\left(1, \text{round}\left(0.003\sqrt{h^2 + w^2}\right)\right)$$

with h, w the image height and width in pixels.

Setup

Eight Cityscapes-pretrained models (MMSegmentation and Hugging Face checkpoints), no fine-tuning, single-scale inference, batch size 1. NVIDIA RTX A5000 (24 GB), Intel Core i9-10940X, 62 GB RAM, Ubuntu 22.04, PyTorch 2.5.1 (CUDA 12.1).